

Automatic Hand-Wash Timer

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According to the World Health Organization (WHO) guidelines, people need to wash their hands with soap and water at least for 20 seconds to get rid of any virus. Although you have a stopwatch on phones, wristwatches, and other electronic devices, it is not convenient to use them. Also, some people do not

want to use their expensive phones near washbasins. So, it becomes difficult to keep a check on time for the handwashing procedure.

Here is a low-cost and simple solution that uses a timer at the heart of the circuit. NE555 timer is used in monostable mode along with an IR sensor to trigger it. The FC-51 IR sensor module has an inbuilt transmitter and receiver.

When you stand in front of the washbasin and place your

hands near the IR sensor module, the sensor detects your hands. The detection makes the sensor to send a low pulse signal to pin 2 of the NE555 timer to trigger it. This activates NE555, and its output pin 3 goes high for 20 seconds. The flush LED1 connected to pin 3 of NE555 glows for 20 seconds. During this time, you can wash your hands until LED1 turns off. Also, a buzzer can be connected to pin 3 of IC1 for audio indication.

The whole circuit can be assem-

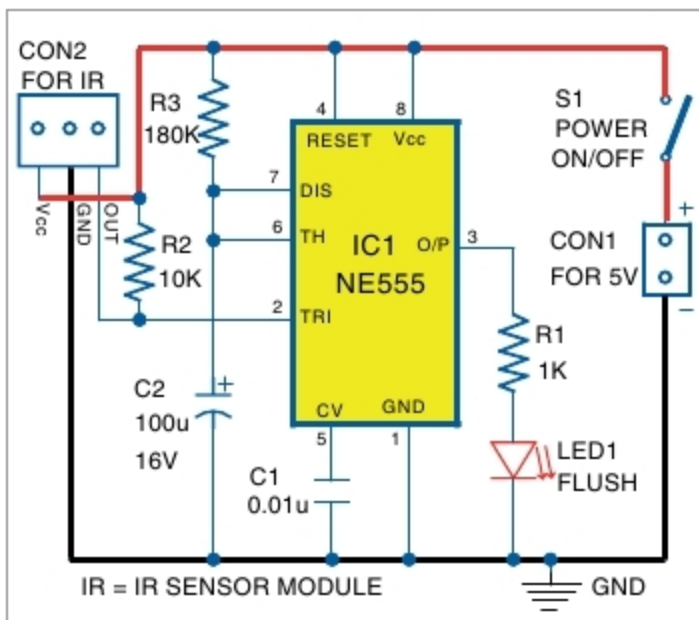


Fig. 1: Circuit diagram of automatic hand-wash timer

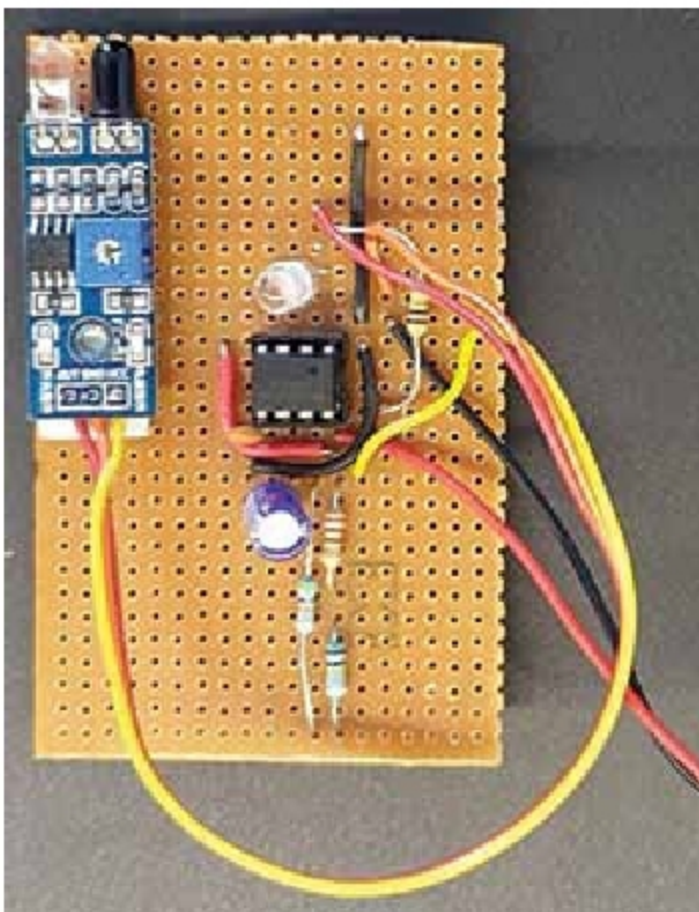


Fig. 2: Author's prototype



Fig. 3: Prototype kept near the washbasin during hand-wash

PARTS LIST

Semiconductors:	
IC1	- NE555 timer
LED1	- 5mm LED
IR	- 3-pin FC-51 IR sensor module
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2	- 10-kilo-ohm
R3	- 180-kilo-ohm
Capacitors:	
C1	- 0.01µF ceramic
C2	- 100µF, 16V electrolytic
Miscellaneous:	
CON1	- 2-pin male connector
CON2	- 3-pin female connector
S1	- On/off toggle switch
	- 5V DC power supply
	- 8-pin IC base

bled on a general-purpose PCB or veroboard. The author's prototype assembled on the veroboard is shown in Fig. 2. The prototype kept in a glass near the washbasin during hand-wash is shown in Fig. 3. **EFY**



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